VI Group Rings

1 The Units Conjecture

Let K be a field, G a group. Let $K[G]$ denote the group algebra with coefficients in K.

The **Units Conjecture** proposed the following: *If K is a field and G is a torsion-free group, then all units in $K[G]$ are trivial.* It was proven false in 2021 by Giles Gardam.

Exercise 1.1

Let $V = \mathbb{Z}_2 \times \mathbb{Z}_2$ be the Klein 4-group. Show that the group algebra $\mathbb{Z}_2[V]$ has a nontrivial unit.

Solution.

The element $1 \cdot (1,1)$ is a unit, with itself as its own inverse. To see this, we can write it as:

$$1 \cdot (1,1) \;=\; 0 \cdot (0,0) + 0 \cdot (1,0) + 0 \cdot (0,1) + 1 \cdot (1,1)$$

Squaring this expression gives our result by using the intuitive multiplication.

Exercise 1.2

Let G be a torsion-free group. Prove that if $K[G]$ has only trivial units, then it has no nonzero nilpotents.

Proof.

Suppose $a^n = 0$ for some $a \in K[G]$. Then, notice:

$$(1-a)(a+a^2+\cdots+a^{n-1}) \;=\; 1-a^n \;=\; 1$$

Thus, $1-a$ is a unit. Since all units of $K[G]$ are trivial, then $1-a = kg$ for some $k \in K, g \in G$. But if $a^n = (1-kg)^n = 0$, then by considering its binomial expansion, we must have that $k = 0$ or $g = e$. If $k = 0$, then $1-a$ is not a unit; if $g = e$, then $a = 0$. ■

Exercise 1.3

Suppose that $K[G]$ has only trivial units. Suppose that $|K|, |G| \geq 3$. Prove that $K[G]$ is semiprimitive.

Proof.

If $a \in J(K[G])$, then $1-a = kg$ is a unit. If $g = e$, then $a = 1-k \in K$, but $a \in J(K[G])$ implies $1 \in J(K[G])$, a contradiction. If $g \neq e$, then since $|K| \geq 3$, pick $c \neq 0, 1$. Then $1-ca = (1-c)+ckg$ is a unit, so necessarily $a = 0$. ■

2 Dietzmann's Lemma

Stating, but not proving.

Lemma 2.1

Let G be a group and let $Z(G)$ be its center. Prove that if $[G : Z(G)] < \infty$, then the commutator subgroup $[G, G]$ is finite.

3 Connell's Theorem - Part I

Let G be a group. Let $\Delta(G)$ be the set of all elements $x \in G$ whose conjugacy class is finite — equivalently, the centralizer $C_G(x)$ has finite index in G — and let $\Delta^+(G)$ be the set of all elements of $\Delta(G)$ of finite order. In this problem we will show that $\Delta^+(G)$ is a subgroup of G.

Exercise 3.1

Let $g_1, \ldots, g_n$ be elements of $\Delta^+(G)$. Let N be the subgroup generated by all the g_i's and their conjugates. Show that N is a finite normal subgroup of G.

Hint: Use Dietzmann's lemma.

Exercise 3.2

Prove that $\Delta^+(G)$ is a subgroup of G.

Exercise 3.3

Prove that G has no finite normal subgroups if and only if $\Delta^+(G) = \{1\}$.

4 Connell's Theorem - Part II

Our goal is to show that if $\Delta^+(G) = \{1\}$, then $K[G]$ is a prime ring. Here we show that $K[\Delta(G)]$ is a prime ring.

Exercise 4.1

Let T be a torsion-free group and assume that every element of T has finite conjugacy class. Prove that $K[T]$ is a commutative domain.

Exercise 4.2

Suppose that G has no finite normal subgroups. Prove that $K[\Delta(G)]$ is a commutative domain.

Hint: Apply part (a) to $T = \Delta(G)$.

5 Neumann's Lemma

Stating, but not proving.

Lemma 5.1

Let G be a group with subgroups $H_1, \ldots, H_m$. Suppose that G can be expressed as a finite union of cosets of the H_i's; that is:

$$G = \bigcup_{i=1}^{m} \bigcup_{j=1}^{n_i} H_i a_{ij}$$

Prove that at least one of the H_i's has finite index in G.

6 Connell's Theorem - Part III

Now we know that $K[\Delta(G)]$ is a (commutative) prime ring. Now we show that this implies that $K[G]$ is also prime.

Let H be a subgroup of G. Then $K[H]$ is a subring of $K[G]$. Define the **restriction map** $\pi_H : K[G] \to K[H]$ by $\pi(\alpha) := \alpha|_H$. For a right ideal I in $K[H]$, let $\rho(I)$ be the right ideal of $K[G]$ generated by I.

Exercise 6.1

Prove that $I \subseteq \rho(\pi(I))$. Thus show that $I = 0$ if and only if $\pi(I) = 0$.

Exercise 6.2

Let I, J be two ideals in $K[G]$. Suppose that $IJ = 0$. Prove that $\pi(I)\pi(J) = 0$.

Hint: This is tricky. Let $x \in I$ and $y \in J$. We must show that $\pi(x)\pi(y) = 0$. First convince yourself it is enough to show that $\pi(x)y = 0$. Now, x can be expressed as a linear combination of elements of G; split this up as $x = x_0 + x_1$ is the part in $K[\Delta(G)]$, and x_1 is the rest (thus $\pi(x) = x_0$). Say $x_0 = \sum_{i=1}^m a_i u_i$ where $u_i \in \Delta(G)$, and $x_1 = \sum_{j=1}^n b_j v_j$, where $v_j \notin \Delta(G)$. Show that the intersection of centralizers $C = \bigcap_{i=1}^m C_G(u_i)$ has finite index in G, and that (if $x_0 y \neq 0$) then it is contained in a union of the cosets of the $C_G(v_i)$'s. Now use Neumann's Lemma.

Exercise 6.3

Let G be any group. Show that if $K[\Delta(G)]$ is prime, then so is $K[G]$.

Exercise 6.4

Deduce Connell's Theorem: that is, if K is a field and G is a group, then $K[G]$ is prime if and only if $\Delta^+(G) = \{1\}$.

7 Faithful Trace Theorem - Part I

A ring R is said to be **Dedekind-finite** if $ab = 1 \implies ba = 1$ for all $a, b \in R$. Our goal is to prove that the group ring $\mathbb{C}[G]$ is always Dedekind-finite, for any group G. To do this, we show that $\mathbb{C}[G]$ possesses a "faithful trace"; from this it is easy to see that every algebra possessing a faithful trace is necessarily Dedekind-finite.

Let R be a $\mathbb{C}$-algebra. A **trace** on R is a linear functional $\tau : R \to \mathbb{C}$ such that $\tau(1) = 1$ and $\tau(ab) = \tau(ba)$ for all $a, b \in R$. A trace τ is said to be **faithful** if $\tau(e) = 0 \implies e = 0$ whenever $e = e^2$ is an idempotent in R.

Exercise 7.1

Prove that if R has a faithful trace, then R is Dedekind-finite.

Proof.

Suppose $ab = 1$ and τ is a faithful trace on R. Then $1 - ba$ is an idempotent:

$$(1 - ba)^2 \;=\; 1 - 2ba + (ba)^2 \;=\; 1 - 2ba + (ba)(ba) \;=\; 1 - 2ba + ba \;=\; 1 - ba$$

It also has zero trace:

$$\tau(1 - ba) \;=\; \tau(1) - \tau(ba) \;=\; \tau(1) - \tau(ab) \;=\; 1 - 1 \;=\; 0$$

Thus, $1 - ba = 0$, so $ba = 1$ as needed. ■

Exercise 7.2

Prove that the normalized trace on $M_n(\mathbb{C})$ is faithful.

Proof.

Recall that $\mathrm{tr}(A)$ is the sum of the eigenvalues of A with multiplicity. If $A^2 = A$, then A is a projection, so $\mathrm{tr}(A) = \mathrm{rank}(A)$, and the result follows. ■

Exercise 7.3

For a group G, define a function τ on $\mathbb{C}[G]$ by $\tau(\alpha) := \frac{1}{|G|}\alpha(1)$. Prove that τ is a trace on $\mathbb{C}[G]$.

ignoring factor cz idt thats generally true. also doesnt say G is finite whats up w that

Proof.

Evidently, $\tau(1) = 1$. We also have:

$$\tau(\alpha\beta) \;=\; \sum_{g\in G}\alpha(g)\beta(g^{-1}) \;=\; \sum_{g\in G}\alpha(g^{-1})\beta(g) \;=\; \tau(\beta\alpha)$$

■

8 Faithful Trace Theorem - Part II

Let G be a group. For an element $\alpha = \sum_{g \in G} \alpha_g g \in \mathbb{C}[G]$, define the **conjugate** of α by:

$$\alpha^* := \sum_{g \in G} \overline{\alpha_g} g^{-1}$$

For $\alpha, \beta \in \mathbb{C}[G]$, define the **inner product** of α with β by:

$$\langle \alpha, \beta \rangle := \tau(\alpha \beta^*)$$

Exercise 8.1

A **self-adjoint idempotent** in $\mathbb{C}[G]$ is an element e such that $e = e^2 = e^*$. Show that if e is a self-adjoint idempotent and $\tau(e) = 0$, then $e = 0$.

This shows that faithfulness holds if e is assumed to be self-adjoint. Thus our goal is to show that every idempotent can be replaced by a self-adjoint one.

Proof.

We have that:

$$\tau(e) \;=\; \tau(ee) \;=\; \tau(ee^*) \;=\; \langle e, e \rangle \;=\; 0$$

Thus, $e = 0$ as needed. ■

Exercise 8.2

Show that $\langle \cdot, \cdot \rangle$ is an inner product. Thus $\mathbb{C}[G]$ is an inner product space.

Exercise 8.3

Let I be a right ideal of $\mathbb{C}[G]$. Show that the orthogonal complement $I^\perp$ is also a right ideal of $\mathbb{C}[G]$.

Proof.

Suppose $b \in I^{\perp}$. We want to show that $bc \in I^{\perp}$ for any $c \in \mathbb{C}[G]$. Note:

$$\langle a, bc \rangle \;=\; \tau(a(bc)^*) \;=\; \tau(ac^*b^*) \;=\; \langle ac^*, b \rangle$$

Since I is a right ideal, $ac^* \in I$ if $a \in I$. Since $b \in I^{\perp}$, then by definition and the above, we get that $\langle a, bc \rangle = 0$, so $bc \in I^{\perp}$ as needed. ■

Exercise 8.4

Let e be an idempotent, and let I be the right ideal generated by e. Let γ be the distance from I to 1, that is:

$$\gamma := \inf\{\|\alpha - 1\| : \alpha \in I\}$$

Prove that $\gamma > 0$.

Proof.

First, note that for any $a = \sum ea_i \in I$, we have:

$$a \;=\; \sum ea_i \;=\; \sum e^2 a_i \;=\; e\left(\sum ea_i\right) \;=\; ea$$

since e is an idempotent. Now, suppose $\gamma = 0$. Then there is a sequence $a_n \in I$ such that $a_n \to 1$. By the above, we also have $a_n = ea_n \to e$, so by uniqueness of limits, $e = 1$. ■

Exercise 8.5

Let γ be as in the previous part. Then, let $(f_n)_{n \geq 1}$ be any sequence in I such that $\|f_n - 1\| \leq \gamma^2 + n^{-4}$.

i) Prove that $|\|f_n\|^2 - \tau(f_n)\| \to 0$ as $n \to \infty$.

ii) Prove that $\|f_n e - e\| \to 0$ as $n \to \infty$.

Exercise 8.6

Finally, show that τ is faithful. Deduce that $\mathbb{C}[G]$ is Dedekind-finite.

(In class I'll show you how to use the $\mathbb{C}$ case to prove a general version that works over any field, in very few lines.)

9 Passmann's Primitivity Theorem

Let K be a field and let G be a group. When is the group ring $K[G]$ primitive? Here we give a necessary condition: if $K[G]$ is primitive and K is an algebraically closed field with $|K| > |G|$, then $\Delta(G) = \{1\}$.

Exercise 9.1

Prove that primitive rings are prime. Thus deduce that if $K[G]$ is primitive, then $\Delta^+(G) = \{1\}$.

Exercise 9.2

Let K be an algebraically closed field, and let R be a primitive K-algebra with $|K| > \dim_K(R)$. Show that $Z(R)$ is 1-dimensional, as follows.

i) Show that $Z(R)$ is a commutative domain.

ii) Let F be the fraction field of $Z(R)$. Show that F/K is a field extension of finite degree. Conclude $F = K$ by algebraic closure.

iii) Conclude $Z(R) = K$. (More precisely, that every element of $Z(R)$ has the form $c \cdot 1$, where $c \in \mathbb{C}$.)

Exercise 9.3

Let C be a finite conjugacy class in G, and let z_C be the sum of elements of C. Prove that the center of the group ring $K[G]$ is spanned by the z_C's.

Exercise 9.4

Now use 2) and 3) [or b) and c)] to prove Passman's Theorem.